Sugar Twin India · system architecture

A digital twin of each Type 2 diabetes patient: fuses EHR (static) with CGM + smartwatch (dynamic) to predict glucose spikes and lows up to 2 hours ahead, and lets a doctor test changes on the patient's own physiology.

1 · DATA (SYNTHETIC, PRIVACY-SAFE)

Synthetic EHR · 5,000 Indian patients

- Demographics, lifestyle, diet (regional staple)
- Diagnoses (ICD-10) incl. CKD, neuropathy
- Labs and vitals, 3 years (LOINC): HbA1c, eGFR…
- Medications: realistic treatment ladder
- Genetics: 4 T2D variants + polygenic score

generate_ehr.py · validated vs ICMR-INDIAB ranges

Wearable + CGM stream · 14 days, 5-min

- CGM glucose (sensor noise, gaps, warm-up)
- Smartwatch: heart rate, HRV, steps, sleep stages
- App logs: meals (≈60% logged), doses, exercise
- Indian meal times, feasts, fasting, illness
- 20 million readings per stream

generate_wearable.py · checked vs real ShanghaiT2DM CGM

Hidden "true" physiology

Insulin sensitivity, beta-cell function, adherence…

drives both streams · never a model input

ShanghaiT2DM (real, open data)

used only to check the simulator is realistic

2 · FUSION

Feature fusion

build_features.py

One row per patient every 30 min

97 features:

CGM trends + personal daily profile
smartwatch + logged meals / doses
EHR labs, meds, genetics

Only what an app knows at that moment

Patient-level split 70 / 15 / 15

3.2 M rows

logged meals + CGM (week 1) → fit each patient's twin

3 · MODELS

LightGBM × 7

Alerts, next 2 h: spike >180 · severe >250 · low <70

Glucose forecast at 30 / 60 / 90 / 120 min

SHAP → plain-language reasons for the doctor

train_lightgbm.py

GRU deep learning (PyTorch)

Raw 6 h sequence (11 channels) + EHR vector

Multi-task: same alerts + forecasts

train_gru.py · GPU

Ensemble

average of LightGBM + GRU (nothing tuned on test)

Personalized physiology twin

Fitted on each patient's week 1 (logged meals + CGM)

Learns: carb sensitivity · recovery time · walk effect

Answers what-if: meal size / type, walk, missed dose

ML forecast + twin's change = scenario curve

twin_simulator.py

4 · SERVING

FastAPI

dashboard/backend/app.py

/patients
risk-sorted triage list

/summary
EHR, genetics, twin, 14 days

/snapshot
forecast, risks, reasons

/whatif
twin scenario simulation

test patients only

5 · DOCTOR DASHBOARD

React + TypeScript

Vite · Tailwind · Recharts

- Patient list sorted by risk (urgent / watch)
- Clinical summary
- 12 h CGM + 2 h forecast with range
- Risks with reasons
- Twin suggestions
- What-if simulator
- Labs, meds, diagnoses, HbA1c trend, genetics
- Demo time machine (replays 14 days)

Light / dark clinical theme

dashboard/frontend

Results on 750 held-out test patients (never used for training)

Spike alert (>180 in next 2 h)	Spikes flagged in advance	Low-sugar alert (<70)	Glucose forecast, 60 min	Twin: new meal's peak	Fusion ablation (spike AUROC)
0.958 AUROC	**95%**	**0.978 AUROC**	**±13.8 mg/dL**	**25 mg/dL error**	**0.952 → 0.956**
0.924 for T2D patients	warning up to 2 h ahead	94.5% of lows flagged early	±17.1 at 120 min	vs 31 one-size-fits-all	CGM only → full fusion
trend rule: 0.71	1.5 alerts per patient-day	severe (>250): 0.968	"no change" baseline: ±21.3	carb sensitivity recovered r = 0.92	60-min error 14.8 → 13.9 mg/dL